

VAPO: Value-Model-Based RL for Advanced Reasoning

Achieves 60.4 on AIME 2024 — Surpassing DAPO by 10+ Points

ByteDance Seed · Qwen2.5-32B base · No SFT data · arXiv:2504.05118

60.4

AIME 2024 avg@32

5,000

Training steps

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO / DAPO)

- Trajectory-level advantage
- Uniform assignment across all tokens
- High variance in credit estimation
- Limited generalization across samples

Value-Model-Based (VAPO)

- Token-level credit assignment
- Precise tracing of each action's impact
- Lower-variance per-token estimates
- Better generalization & sample efficiency

Critical insight: subtle step-level errors cause catastrophic failures in multi-step reasoning — value models catch them.

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

01

Value Model Bias

Long trajectories and bootstrapped learning introduce severe bias in value estimation, destabilizing early training.

02

Heterogeneous Lengths

Different response lengths require different bias-variance tradeoffs — a single λ fails across the board.

03

Reward Sparsity

Only the final `<eos>` token receives reward signal, making credit assignment across thousands of tokens extremely hard.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Value-Pretraining

→ Value Model Bias

Decoupled-GAE

→ Value Model Bias

Length-adaptive GAE

→ Heterogeneous
Lengths

Clip-Higher

→ Exploration /
Stability

Token-level Loss

→ Reward Sparsity

**Positive Example LM
Loss**

→ Reward Sparsity

Group-Sampling

→ Baseline
Estimation

Each component is not an isolated trick — they jointly address the three core failure modes of value-model-based RL in long-CoT.

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Short Responses

Lower $\lambda \rightarrow$ lower variance

Fewer tokens mean bootstrapped estimates are more reliable; bias is already low.

Long Responses

Higher $\lambda \rightarrow$ lower bias

Many tokens accumulate bootstrap error; raising λ leans more on Monte-Carlo returns to reduce bias.

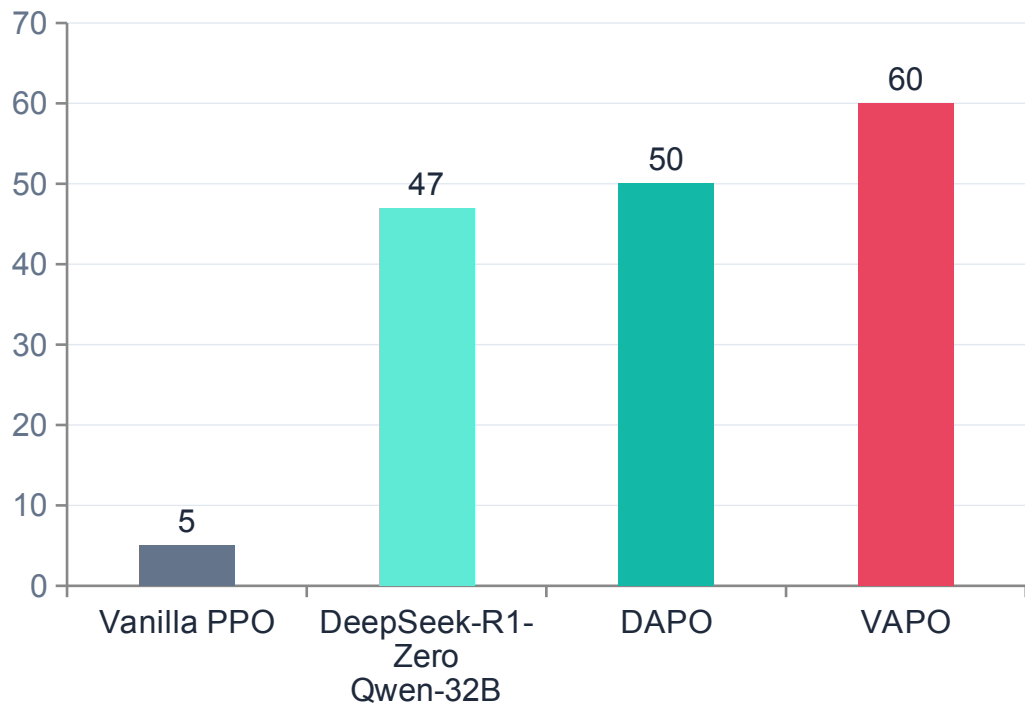
Why It Matters

Standard GAE uses a fixed λ for all sequences. In long-CoT reasoning, response lengths vary wildly (hundreds to thousands of tokens). A single λ cannot simultaneously minimize bias for long chains and variance for short chains.

VAPO's Length-adaptive GAE dynamically sets λ per response, optimizing the bias-variance tradeoff across the full length distribution.

Contribution: +15 points (removing Length-adaptive GAE drops AIME score from 60 $\rightarrow$ 45)

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



+10.4

points over DAPO

+13

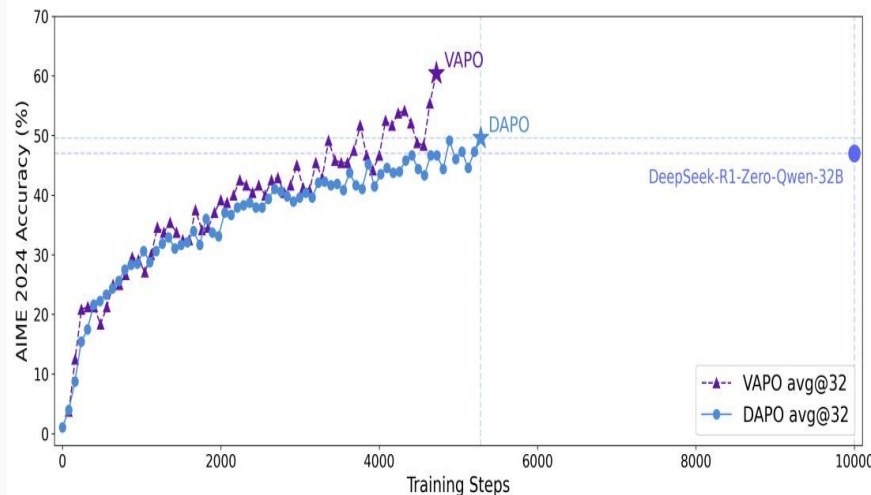
points over DeepSeek-R1-Zero

Same Base

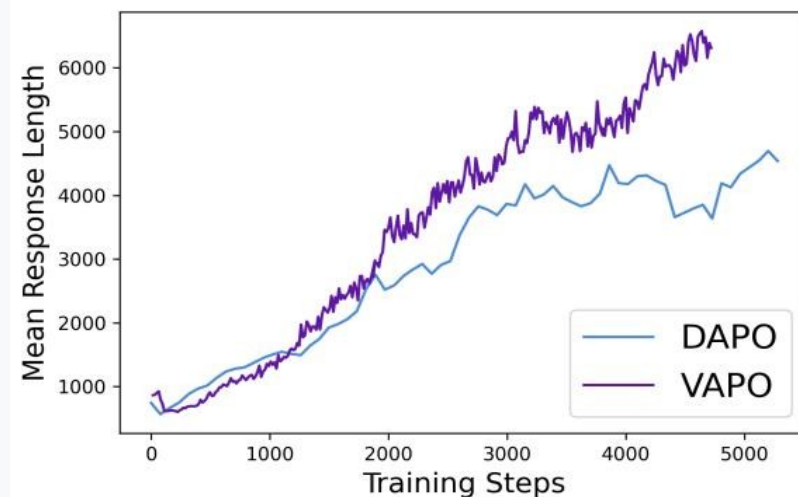
Qwen2.5-32B, no SFT

Training curve: VAPO consistently outperforms DAPO throughout training (see next slide).

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO



Training Accuracy Curve (AIME 2024)



Mean Response Length Over Training

**VAPO reaches
60%**

at step ~4,500

**DAPO plateaus
~50%**

at step ~5,500

Zero crashes

across multiple runs

~6,000 tokens

VAPO avg response length

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Configuration	AIME 2024 avg@32	Δ from VAPO
Vanilla PPO	5	-55
VAPO w/o Value-Pretraining	11	-49
VAPO w/o Decoupled-GAE	33	-27
VAPO w/o Length-adaptive GAE	45	-15
VAPO w/o Clip-Higher	46	-14
VAPO w/o Token-level Loss	53	-7
VAPO w/o Positive Example LM Loss	54	-6
VAPO w/o Group-Sampling	55	-5
VAPO (full)	60	—

+49

Value-Pretraining

60 → 11 without it

+27

Decoupled-GAE

60 → 33 without it

+15

Length-adaptive GAE

60 → 45 without it

**+5—
14**

Other components

Collective gain

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1

Value-model-based methods can surpass value-free when core challenges are systematically addressed.

2

Length-adaptive GAE is critical for heterogeneous sequence lengths, contributing 15 points.

3

Value pretraining is the single most important component — without it, score collapses from 60 to 11.

4

VAPo is highly stable: zero training crashes across multiple independent runs.

5

Implication: future reasoning models should reconsider value-based RL as a first-class citizen, not a legacy technique.